

V: Potential Energy: $V(x) = - \int F(x) dx$

• The constant the integral gives doesn't matter: we can choose its value.

• Conservative forces: $F = - \frac{dV}{dx}$

total Energy: $E = T + V$

$\uparrow \quad \uparrow$
 Kinetic Energy Potential Energy
 $T = \left(\frac{1}{2} m v^2 \right)$

Back to Circular motion:

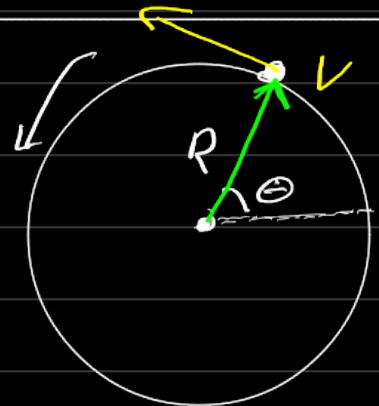
• ω : angular frequency

$$v = \frac{dx}{dt}$$

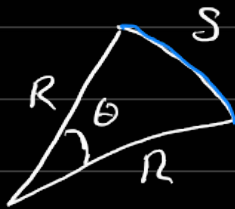
$$\omega = \frac{d\theta}{dt}$$

$$\omega = \frac{v}{R}$$

(Magnitude of V)



Now



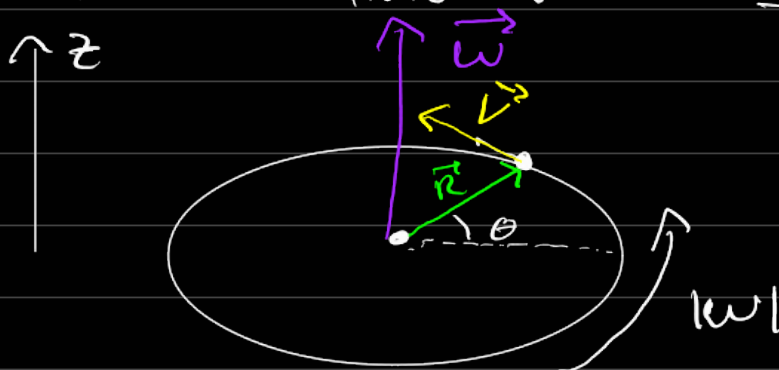
$$S = \theta R$$

$$\frac{dS}{dt} = \frac{d\theta}{dt} R$$

(w)

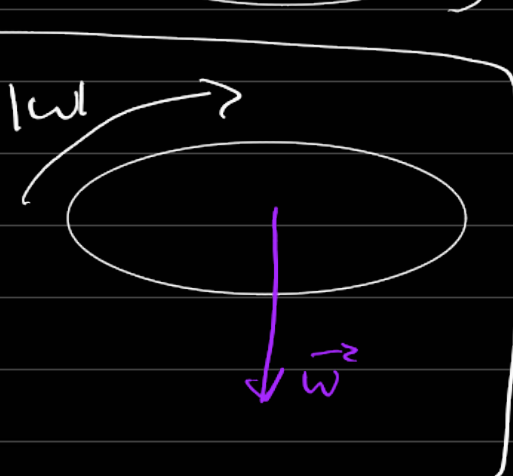
$$v = \frac{d\theta}{dt} R \Rightarrow \frac{d\theta}{dt} = \frac{v}{R}$$

* Turn these into vector equations:



$\vec{\omega}$: Normal to the surface rotating

$$\omega = \frac{v}{R} *$$



$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{\omega} \times \vec{r} = - \vec{r} \times \vec{\omega}$$

$$\begin{cases} \vec{A}_1 = \vec{A}_2 \times \vec{A}_3 \\ \vec{V} = \vec{\omega} \times \vec{R} \end{cases}$$

$$|\vec{V}| = |\vec{\omega} \times \vec{R}| = |\vec{\omega}| |\vec{R}| \sin(\theta) \\ = |\vec{\omega}| |\vec{R}| *$$

$\theta = 90$

"regular" momentum

$$p = mV$$

$$\vec{p} = m\vec{V}$$

(linear momentum)

New:

• Angular Momentum: L

$$\vec{L} = \vec{r} \times \vec{p} *$$

$$\vec{p} = m\vec{V} = m \frac{d\vec{r}}{dt} = m\dot{\vec{r}}$$

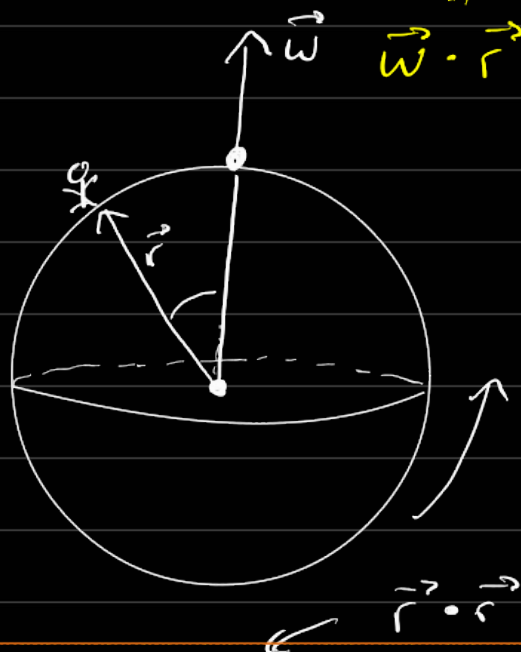
$$\vec{V} = \dot{\vec{r}}$$

$$(i) \vec{L} = m \vec{r} \times \dot{\vec{r}}$$

$$\begin{aligned} (ii) \vec{L} &= \vec{r} \times \vec{p} = m \vec{r} \times \vec{V} \\ &= m \vec{r} \times (\vec{\omega} \times \vec{r}) \\ &= m (\vec{\omega} (\vec{r} \cdot \vec{r}) - \vec{r} (\vec{\omega} \cdot \vec{r})) \end{aligned}$$

$$\vec{V} = \vec{\omega} \times \vec{r}$$

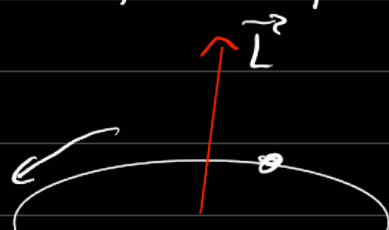
Above we used R but the notes use r .
→ Go over to r



$$\vec{L} = m [\vec{\omega} |\vec{r}|^2 - \vec{r} (\vec{\omega} \cdot \vec{r})] \quad (iii)$$

* Look at the ang. momentum for the case we did before
 $\vec{\omega} = \omega \hat{e}_z$, $\vec{\omega} \cdot \vec{r} = 0$ (set up the problem)

$$\vec{L} = m \vec{\omega} |\vec{r}|^2 = m \omega |\vec{r}|^2 \hat{e}_z$$



$$|\vec{L}| = m\omega \underbrace{|\vec{r}|^2}_{R^2}$$

$$|\vec{r}|^2 = R^2$$

*

$$|\vec{L}| = m\omega R^2 \quad (\text{Uniform Circular motion})$$

$$\hookrightarrow V = \omega R$$

• "get rid of ω " : $\omega = \frac{V}{R}$

$$|\vec{L}| = m\left(\frac{V}{R}\right)R^2 = mVR$$

$$|\vec{L}| = mVR$$

• "get rid of R " : $R = \frac{V}{\omega}$

$$|\vec{L}| = m\omega \left(\frac{V}{\omega}\right)^2 = m \frac{V^2}{\omega}$$

$$|\vec{L}| = m \frac{V^2}{\omega}$$

Claim: $\vec{L}$ is conserved. Recall for energies this means

$$\hookrightarrow E_i = E_f$$

$$E_f - E_i = 0$$

$$\Delta E = E_f - E_i = 0$$

divide by a small time

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\Delta E}{\Delta t} \right) = 0$$

$$\Delta t = t_2 - t_1$$

$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta f}{\Delta x} \right)$$

$$= \lim_{\Delta x \rightarrow 0} \left(\frac{f(x + \Delta x) - f(x)}{(x_2 - x_1)} \right)$$

$$\frac{dE}{dt} = 0$$

(Conservation of energy)

* Likewise, we can have other conserved quantities

$$\frac{dQ}{dt} = 0 \Rightarrow Q : \text{conserved}$$

* $\vec{L}$ is conserved : $\frac{d\vec{L}}{dt} = 0$

$$\vec{L} = \vec{r} \times \vec{p} \Rightarrow \frac{d\vec{L}}{dt} = \frac{d}{dt} (\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$$

$$\begin{aligned} \left[\begin{array}{ccc} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ r_x & r_y & r_z \\ p_x & p_y & p_z \end{array} \right] &= (r_x p_z - r_z p_x) \hat{e}_x - (\dots) + (\dots) \\ \frac{d}{dt} (\dots) &= \frac{d}{dt} (r_x p_z - r_z p_x) \hat{e}_x \\ &= \left(\frac{dr_x}{dt} p_z + r_x \frac{dp_z}{dt} - \left(\frac{dr_z}{dt} p_x + r_z \frac{dp_x}{dt} \right) \right) \hat{e}_x \end{aligned}$$

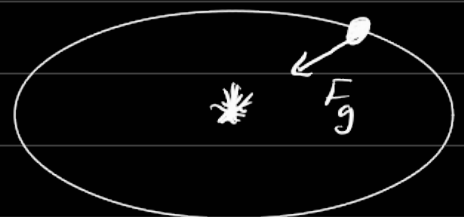
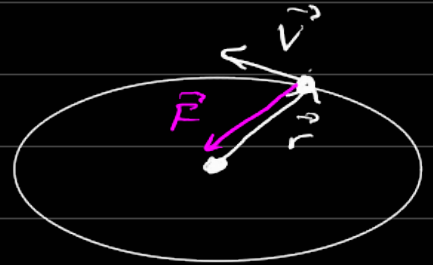
$$\textcircled{1} \quad \vec{r} \times \vec{p} = \vec{r} \times (m \vec{v}) = \vec{r} \times (m \dot{\vec{r}}) = m (\vec{r} \times \dot{\vec{r}}) = 0$$

$\vec{a} = \dot{\vec{v}}$

$$\begin{aligned} \textcircled{2} \quad \vec{r} \times \frac{d\vec{p}}{dt} &= \vec{r} \times \frac{d}{dt} (m \vec{v}) = \vec{r} \times (m \vec{a}) \\ &= \vec{r} \times (m \vec{a}) = \vec{r} \times \vec{F} = 0 \end{aligned}$$

$$\boxed{\frac{d\vec{L}}{dt} = 0}$$

(Conserved for Central forces)



* Strictly speaking this is true when F is a central force

